Ryan Bahlous-Boldi ryanbb@mit.edu Website Scholar

EDUCATION

Massachusetts Institute of Technology Cambridge, MA
Ph.D. in Electrical Engineering and Computer Science Sep 2025 – Present

University of Massachusetts Amherst Amherst, MA
B.S. in Computer Science, *summa cum laude* Sep 2021 – May 2025
GPA: 4.0, Member of Commonwealth Honors College
Minors: Philosophy, Psychology

RESEARCH INTERESTS

Reinforcement Learning · Continual Learning · Diversity & Exploration · Open-Endedness · Evolutionary Computation

RESEARCH EXPERIENCE

Improbable AI Lab 9/2025 – Present
Massachusetts Institute of Technology Cambridge, MA
Advisor: Pulkit Agrawal
- Studying reinforcement learning and continual learning, including training methods that preserve diversity to improve exploration and test-time search.

Flowers Team, INRIA Bordeaux 3/2024 – 11/2024
Advisors: Clément Romac, Cédric Colas, Pierre-Yves Oudeyer

PUSH Lab, UMass Amherst & Amherst College 9/2021 – 5/2025
Advisor: Lee Spector. Applied lexicase selection to genetic programming and evolutionary reinforcement learning, with a focus on exploration and diversity.

SCALAR Lab, UMass Amherst 9/2022 – 5/2025
Advisor: Scott Niekum. Improved safety and alignment of RLHF systems by learning diverse reward functions and policies from preferences with hidden context.

AART Lab, Carnegie Mellon University 5/2024 – 12/2024
Advisor: Katia Sycara

ICAROS Lab, University of Southern California 5/2023 – 3/2024
Advisor: Stefanos Nikolaidis

PUBLICATIONS

* denotes equal contribution

Preprints

Jyothish Pari, **Ryan Bahlous-Boldi**, and Pulkit Agrawal. 2026. Dynamic Compression in Recurrent Networks. Posted at https://arxiv.org/abs/2608.17896.

Ryan Bahlous-Boldi, Isha Puri, Idan Shenfeld, Akarsh Kumar, Mehul Damani, Sebastian Risi, Omar Khattab, Zhang-Wei Hong, and Pulkit Agrawal. 2026. Vector Policy Optimization: Training for Diversity Improves Test-Time Search. Posted at https://arxiv.org/abs/2605.22817.

Journal Articles

Ryan Boldi*, Martin Briesch*, Dominik Sobania, Alexander Lalejini, Thomas Helmuth, Franz Rothlauf, Charles Ofria, and Lee Spector. 2023. Informed Down-Sampled Lexicase Selection: Identifying productive training cases for efficient problem solving. https://arxiv.org/abs/2301.01488. In *Evolutionary Computation*. MIT Press.

Book Chapters

Lee Spector, **Ryan Bahlous-Boldi**, Paul Vander Vort, and Niko Lorantos. 2025. Evolution of Artificial Intelligence, Continued. In Genetic Programming Theory and Practice XXII. New York: Springer. To appear.

Lee Spector, Li Ding and **Ryan Boldi**. 2024. Particularity. In Genetic Programming Theory and Practice XX. New York: Springer. https://doi.org/10.1007/978-981-99-8413-8_9. Preprint posted at https://arxiv.org/abs/2306.06812

Conference Papers

Akarsh Kumar, **Ryan Bahlous-Boldi**, Prafull Sharma, Phillip Isola, Sebastian Risi, Yujin Tang, and David Ha. 2026. Digital Red Queen: Adversarial Program Evolution in Core War with LLMs. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26). Posted at https://arxiv.org/abs/2601.03335.

Ryan Bahlous-Boldi, Li Ding, Lee Spector, and Scott Niekum. 2025. Pareto-Optimal Learning from Preferences with Hidden Context. In the Reinforcement Learning Conference 2025. Posted at https://arxiv.org/abs/2406.15599.

Ryan Bahlous-Boldi*, Maxence Faldor*, Luca Grillotti, Hannah Janmohamed, Lisa Coiffard, Lee Spector, and Antoine Cully. 2025. Dominated Novelty Search: Rethinking Local Competition in Quality-Diversity. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '25). Posted at https://arxiv.org/abs/2502.00593

Ryan Boldi, Ashley Bao, Martin Briesch, Thomas Helmuth, Dominik Sobania, Lee Spector, Alexander Lalejini. 2024. Untangling the Effects of Down-Sampling and Selection in Genetic Programming. In Proceedings of ALIFE 2024: the 2024 Conference on Artificial Life. (ALIFE '24). https://doi.org/10.1162/isal_a_00832. MIT Press. Posted at https://arxiv.org/abs/2304.07089.

Ryan Boldi and Lee Spector. 2023. Can the Problem-Solving Benefits of Quality Diversity Be Obtained Without Explicit Diversity Maintenance? In Genetic and Evolutionary Computation Conference Companion (GECCO '23).

Li Ding, **Ryan Boldi**, Thomas Helmuth, and Lee Spector. 2022. Lexicase selection at scale. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22).

Workshop Papers

Ryan Boldi, Li Ding and Lee Spector. 2023. Objectives Are All You Need: Solving Deceptive Problems Without Explicit Diversity Maintenance. In the Workshop on Agent Learning in Open-Endedness at NeurIPS (ALOE @ NeurIPS 2023). Posted at https://arxiv.org/pdf/2311.02283

Ryan Boldi, Thomas Helmuth, and Lee Spector. 2022. The environmental discontinuity hypothesis for down-sampled lexicase selection. In The 2022 Conference on Artificial Life - Why it Didn't Work-Shop (ALIFE '22). Posted at https://arxiv.org/pdf/2205.15931

Posters and Poster Papers

Ryan Boldi, Li Ding and Lee Spector. 2023. Solving Deceptive Problems without Explicit Diversity Maintenance. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24).

Ryan Boldi, Matthew Fontaine, Sumeet Batra, Gaurav Sukhatme and Stefanos Nikolaidis. 2024. Generating Diverse Induced Policies for Conditioned Policy Distillation. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '24)

Ryan Boldi, Charles Zhang, Lee Spector. 2023. Encouraging Diversity in Reinforcement Learning with Lexicase Selection. RL at Harvard Workshop 2023.

Ryan Boldi, Ashley Bao, Martin Briesch, Thomas Helmuth, Dominik Sobania, Lee Spector, Alexander Lalejini. 2023. The Problem Solving Benefits of Down-Sampling Vary by Selection Scheme. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '23).

Ryan Boldi, Alexander Lalejini, Thomas Helmuth, Lee Spector. 2023. A static analysis of informed down-samples. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '23).

Li Ding, **Ryan Boldi**, Thomas Helmuth, and Lee Spector. 2022. Going faster and hence further with lexicase selection. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '22).

TEACHING AND MENTORSHIP

Teacher & Teaching Research Team, *X-Camp Academy* — 9/2021–Present

- Mentored 500+ students in USA Computing Olympiad (USACO) preparation; designed and taught X-Camp's inaugural AI curriculum, scaling it from 5 to 30+ students per cycle.
- Led migration of academic operations (15+ instructors, 500+ students) to a new ed-tech platform; manage teacher recruitment, interviewing, and onboarding.

President, *MassAI* - UMass Amherst's AI Student Organization — 1/2023–1/2025
Team Leader, *Team UMass: ProjectX ML Research Comp. Winners* — 9/2022–1/2023

PRESENTATION

(*In addition to those listed as conference/workshop papers and posters above*)

Conference

Encouraging Diversity in Reinforcement Learning with Lexicase Selection
Poster: RL at Harvard Workshop 2023 — Cambridge, MA

Think Before You Act: Generating High-Quality Diverse Reasoning Policies
Poster: SoCal Undergraduate Research Symposium 2023 — Los Angeles, CA

The Emergence of Diversity
Emerging Researchers in Artificial Life Lightning Talk
2023 Conference on Artificial Life — Sapporo, Japan

Invited

Evolutionary Computation — Spring 2023
UMass Amherst Guest Lecture — Amherst, MA
COMPSCI 389 - Introduction to Machine Learning

Lexicase Selection and Reinforcement Learning — Fall 2022
Personal Autonomous Robotics Lab (PeARL), UT Austin — Austin, Texas
Autonomous Learning Laboratory, UMass Amherst — Amherst, MA

Lexicase Selection and the Diversity of Quality — Summer 2022

	Adaptive and Intelligent Robotics Lab, Imperial College London	London, UK
AWARDS	*NSF Graduate Research Fellowship (GRFP)* National Science Foundation, 2025–2030	$159,000
	21^{st} *Century Leaders Award* University of Massachusetts Amherst, 2025	
	Outstanding Undergraduate Achievement Award Manning College of Information and Computer Sciences, 2025	
	Selected as an *HRI Pioneer* HRI'25: IEEE/ACM International Conference on Human-Robot Interaction	
	Goldwater Scholarship Barry Goldwater Scholarship & Excellence in Education Foundation, 2024	$7,500
	ProjectX ML Research Competition Winner University of Toronto, 2023	$20,000
MEMBERSHIP	*International Society for Artificial Life* *ACM SIGEVO, Special Interest Group for Genetic and Evolutionary Computation* *IEEE Robotics & Automation Society*	
SERVICE	*Reviewer, NeurIPS 2026* *Reviewer, ALIFE 2025* *Reviewer, ICLR 2025* *Volunteer, RLC 2024* *Reviewer, GECCO 2024 Graph-based GP Workshop* *Volunteer, GECCO 2023*	
COMPUTER SKILLS	*Languages & Frameworks* Python, Clojure, C++, Java, JavaScript, R, Numpy, PyTorch, Jax, Flax	